

Nanjing Qineng Electronic Technology Co., Ltd.

QNDBK3 Series Split Core Single-Phase Current Sensor Instruction Manual:

1、Product Name: Split Core Single-Phase Current Sensor

2、Product Specifications: QNDBK3/RS485

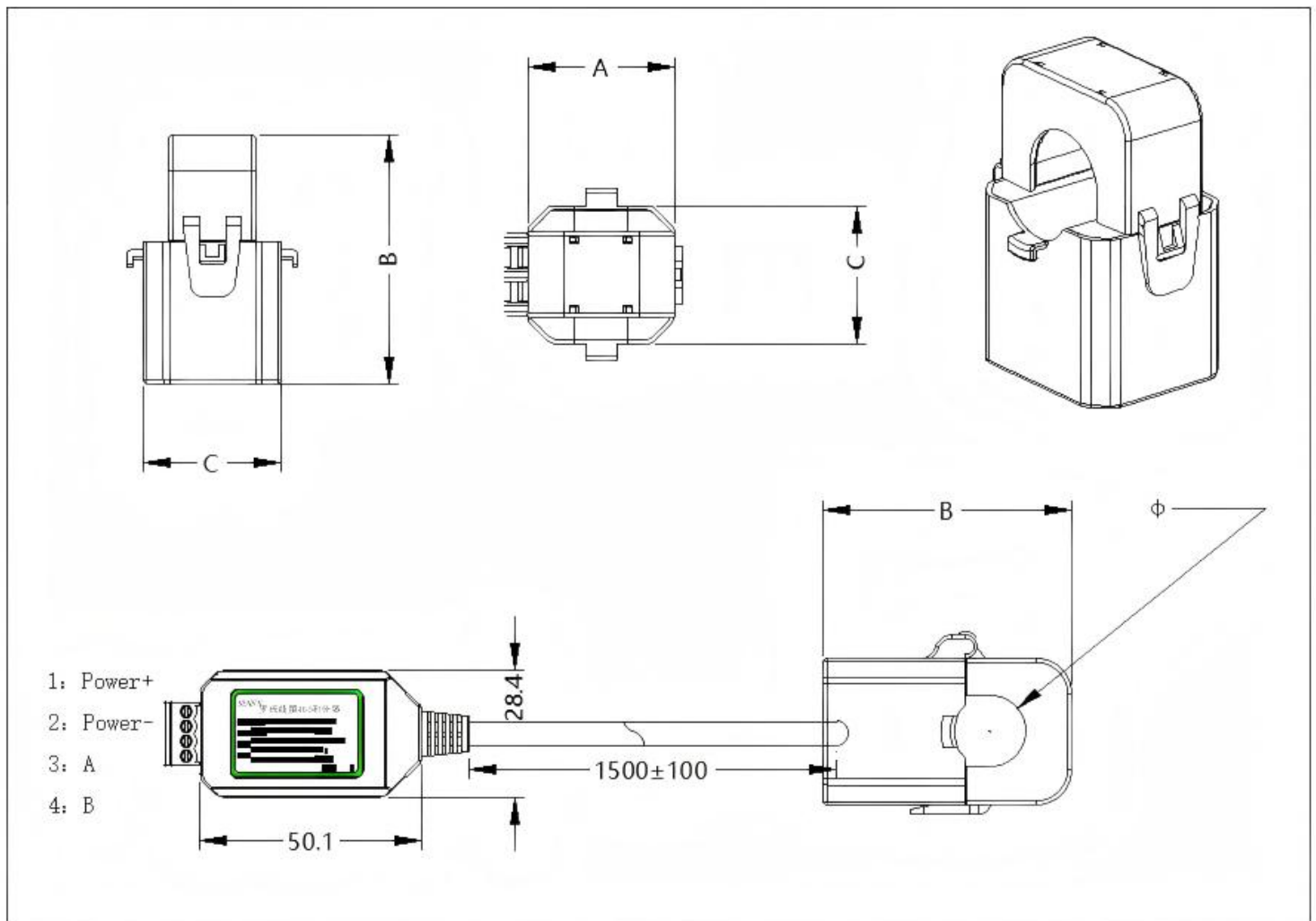
3、Overview: This product is a switchable single-phase AC digital current transformer. It utilizes a professional MCU for high-precision measurement of the effective value of single-phase AC current, outputting an RS-485 signal. It is suitable for power grid monitoring, motor signal detection, railway system signal detection, etc. Internally, it employs advanced surface mount technology, ensuring long-term stability and strong anti-interference capabilities. The input, output, and power supply terminals can withstand high surge voltage impacts. Its compact and elegant design makes it suitable for integrated installation.

Technical parameters(50Hz), Special parameters can be customized:

Rated current	5A, 10A, 20A, 50A, 63A, 100A, 200A, 250A, 300A, 400A, 500A, 630A, 800A, 1000A, 1500A, 2000A(optional)
Rated Output	RS485
Power Supply	DC 12V
10%-100% Linear	±0.5%
Inner Diameter	10mm/16mm/24mm/36mm/50mm(optional)
Use Altitude	W2000M
Magnetic Core Material	Ferrite/Silicon Steel
Shell Material	Nylon
Secondary Lead	2 x 0.3mm ² 1.5m sheathed cable (length optional)
Insulation Resistance	DC500V/100MQ
Usage Conditions	-40℃~75℃
Storage Conditions	-30℃~75℃
Accuracy Level	For fixed-position installation, the tolerance is 0.5; for non-fixed-position installation, the tolerance is 1.5%.

Product Dimension Drawing: Unit (mm) Tolerance ± 1

Model	Rated Input	External dimensions			
		aperture	A	B	C
QNDBK3-10	5-75A	10mm	23	40	27
QNDBK3-16	5-160A	16mm	31	50	30
QNDBK3-24	10-300A	24mm	43	73	39
QNDBK3-35	10-630A	35mm	60	89	43
QNDBK3-50	10-2000A	50mm	81.2	119.8	58



1、Protocol Frame Format:

Definition	Address code	Function code	Data area	CRC check
Byte	1	1	N	2
Illustrate:	1: Address code: Values range from 1 to 250; 0xFF indicates a broadcast frame. 2: Function code: 03H reads register value, 10H writes register value. 3: Data area: When reading a register, it indicates the register's starting address and the number of records to be read; when writing a register, it indicates the register's starting address, the number of records to be written, and the corresponding value. 4: CRC checksum: 16-bit CRC checksum, with the least significant byte first and the most significant byte last. 5: Both register address and register quantity are in 2-byte format.			

2、Example of reading a register:

Host reads registers 1000H-1003H			Slave return registers 1000H-1003 data		
Host sends		Send message	Slave response	Return information	
Address code		01H	Address code	01H	
Function code		03H	Function code	03H	
Starting address	high byte	10H	Data bytes	08H	
	low byte	00H		08H	
Number of registers	high byte	00H	1000H register data	high byte	00H
	low byte	04H		low byte	0BH
CRC check	low byte	40H	1001H register data	high byte	00H
	high byte	C9H		low byte	00H
			1002H register data	high byte	00H
				low byte	00H
			1003H register data	high byte	00H
				low byte	00H
			CRC check	low byte	2FH
				high byte	17H

3、Example of writing to a register:

Host writes data to registers 101AH-101CH			The slave device returned response data	
Host sends		Send message	Slave response	Return information
Host sends		01H	Address code	01H
Address code		10H	Function code	10H
Starting address	high byte	10H	Starting address	10H
	low byte	1AH		1AH
Number of registers	high byte	00H	Number of registers	00H
	low byte	03H		03H
Data bytes		06H	CRC check	05H
Write data to 101AH register	high byte	01H		0FH
	low byte	2CH		
Write data to register 101BH	high byte	00H		
	low byte	3CH		
Write data to register 101CH	high byte	00H		
	low byte	3CH		
CRC check	low byte	C3H		
	high byte	FFH		

4、Register Description:

Address	Parameter definition	Reading and writing	illustrate
1002H	The least significant bit of the current sample value includes one decimal place.	R	Current value range: 0-60000A
100BH	Device Communication ID	R/W	Communication address, value range 1-250
100CH	Device communication baud rate	R/W	The default baud rate is 9600bps.

Example: Baud rate 4800 Serial port transmit: FF 10 10 0C 00 01 02 00 00 FF 39 Serial port receive: 01 10 10 0C 00 01 C5 0A

Explanation: 00 00 corresponds to the baud rate value. Communication baud rates range from 0-4800, 1-9600, 2-19200, and 3-38400.

FF 39 represents the CRC16 checksum for a baud rate of 4800.

Instructions for reading data from the host computer (RS485 query software)	
01 03 10 01 00 02	Low-order bit read/retrieve value 0A~60000A
FF 10 10 0B 00 01 02 00 01	Modify the address; the last four digits will change (in hexadecimal).
Example CRC16: (01 03 10 01 00 02) 91 0B	91 0B is the CRC16 checksum of address 1
A CRC16 checksum must be added	
Host computer data parsing	
Host transmits read register protocol data structure format analysis	01 03 10 01 00 02 91 0B
Device address 1 byte	01 indicates the device address
Read register command 1 byte	03 indicates a register read instruction
Register address 2 bytes	10 01 represents the starting address of the register
Number of registers read 2 bytes	00 02 indicates reading two registers
CRC checksum (high byte first, low byte last) 2 bytes	91 0B indicates the CRC checksum
For example:	
The slave return register is a data structure parsing.	01 03 04 00 00 14 F3 B5 76
Device address 1 byte	01 indicates the device address
Read register command 1 byte	03 indicates a register read instruction

Number of bytes read N bytes	04 indicates that there are four hexadecimal data
XX * N Nbytes of data read	14 F3 indicates valid sensor data (four hexadecimal digits: 00 00 14 F3)
CRC check (high byte first, low byte last)	B5 76 indicates the CRC checksum

5、Register read/write instructions:

① MODBUS data frames are represented in hexadecimal format. For example, address code 10H corresponds to decimal address 16, and function code 10H also corresponds to decimal address 16, indicating data write. Each MODBUS register corresponds to 16 bits of data, with a value from 0 to 65535, represented in hexadecimal as 0000H to FFFFH, or 0x0000 to 0xFFFF.

② R indicates read-only, R/W indicates read-write, and W indicates write-only; the current calibration value corresponds to registers 1003H and 1004H, representing the high and low bits of data respectively, which are read-only and cannot be written; the device ID corresponds to register 100BH, which can be read and written to modify the device ID; the calibration current corresponds to register 1010H, which can only write the calibration current value and cannot be read.

③ When reading data from a MODBUS frame, function code 03H is used, and 16 registers from 1000H to 100FH can be read: To read all 16 registers consecutively, the general instruction is FF 03 10 00 00 10 55 18, where FFH represents the broadcast frame address 255, 03H represents the read function, 1000H represents the start address, 0010H represents the number of registers 16, and 5518H represents the checksum, with the least significant byte first and the most significant byte last; to read only the device ID, the general instruction is FF 03 10 0B 00 01 E4 D6, where FFH represents the broadcast frame address 255, 03H represents the read function, 100BH represents the start address, 0001H represents the number of registers 1, and E4D6H represents the checksum, with the least significant byte first and the most significant byte last.

④ When writing data to a MODBUS frame, function code 10H is used, and 12 registers (1006H~100BH) can be written. To modify the output type and coil type, the general instruction is FF 10 10 08 00 02 04 00 04 00 0B 08 24, where FFH represents the broadcast frame address 255, 10H represents the write function, 1008H represents the start address, 0002H represents the number of registers (2), 04H represents the subsequent data length (4 bytes), 0004H indicates that the data in register 1008H has been modified to 4, corresponding to an output type of 4~20mA, 000BH indicates that the data in register 1009H has been modified to 11, corresponding to a coil type of 1000A/100mV, and 0824H represents the checksum, with the least significant

byte first and the most significant byte last. To modify the device ID, the general instruction is FF 10 10 0B 00 01. 02 00 02 7F 4F, where FFH represents broadcast frame address 255, 10H represents write function, 100BH represents start address, 0001H represents the number of registers 1, 02H represents the length of subsequent data 2 bytes, 0002H represents register 100BH data modified to 2, corresponding to device ID 2, 4F7FH represents checksum, with low byte first and high byte last; For calibration current, the general instruction is FF 10 10 10 00 01 02 00 C8 FC F3, where FFH represents broadcast frame address 255, 10H represents write function, 1010H represents start address, 0001H represents the number of registers 1, 02H represents the length of subsequent data 2 bytes, 00C8H represents register 1010H data modified to 200, corresponding to calibration current value 200A, FCF3H represents checksum, with low byte first and high byte last;

⑤ 03 Function code, communication rate 9600bps, 8 data bits, 1 stop bit, no parity, 4-byte integer unit 0.01A, using RTU mode in MODBUS communication protocol.

⑥ The factory default address is 01 (can be modified), and the default baud rate is 9600bps. Precision is 0.3%. The algorithm for calculating the actual current value is: hexadecimal data converted to decimal data $\div 10$ = actual current.

6、ModbusCAN Software Data Query Operation Instructions (Essential Reading for PLC/MODBUSCAN/485 Host Computer)

